

RS ASSIGNMENT#02

QUESTION#01

Paper: *Multiview Fusion Using Transformer Model for Recommender Systems: Integrating the Utility Matrix and Textual Sources*

Algorithm: Multiview Transformer Model for Recommendation

Input:

- User–Item Utility Matrix U
- Item Textual Features T
- Model parameters (embeddings, transformer weights)

1. Preprocessing

- Convert utility matrix values into user–item embeddings:
$$E_u = \text{Embed}(U)$$
- Tokenize textual data:
$$X = \text{Tokenizer}(T)$$
- Generate text embeddings:
$$E_t = \text{TextEmbed}(X)$$

2. Multiview Embedding Construction

- Concatenate utility view and text view:
$$H_0 = [E_u \parallel E_t]$$

3. Transformer-Based Fusion

For each transformer layer $L = 1 \dots N$:

a. Multi-Head Self-Attention:

$$A_L = \text{MHSA}(H_{L-1})$$

b. Add & Norm:

$$S_L = \text{LayerNorm}(H_{L-1} + A_L)$$

c. Feed-Forward Network:

$$F_L = \text{FFN}(S_L)$$

d. Final Add & Norm:

$$H_L = \text{LayerNorm}(S_L + F_L)$$

4. Fusion Output

- Pool final transformer output:
$$Z = \text{Pooling}(H_N)$$

5. Prediction Layer

- Final score prediction:
$$\hat{r}_{ui} = \sigma(WZ + b)$$

6. Training

- Optimize using Mean Squared Error:
$$L = \frac{1}{N} \sum (r_{ui} - \hat{r}_{ui})^2$$

7. Output

- Predicted user–item rating or ranked recommendation list

This algorithm integrates utility matrix data and item text via embeddings, processes them through a transformer encoder for multiview fusion, and outputs predicted ratings. It formalizes steps from preprocessing to training, precisely capturing the paper's fusion methodology for recommendation tasks.

QUESTION#02

Paper: *Multiview Fusion Using Transformer Model for Recommender Systems*

Part 1: Mathematical Model

1. Input Representations

Utility Matrix View:

$$E_u = f_u(U) \in \mathbb{R}^d$$

Textual View:

- Tokens:

$$X = \{x_1, x_2, \dots, x_n\}$$

- Embeddings:

$$E_t = \text{Embed}(X) \in \mathbb{R}^{n \times d}$$

2. Joint Embedding

$$H_0 = [E_u \parallel E_t] \in \mathbb{R}^{(n+1) \times d}$$

3. Multi-Head Self-Attention

$$Q = H_{L-1}W_Q, K = H_{L-1}W_K, V = H_{L-1}W_V$$

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

$$\text{MHSA} = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W_O$$

4. Transformer Layer

$$S_L = \text{LayerNorm}(H_{L-1} + \text{MHSA})$$

$$H_L = \text{LayerNorm}(S_L + \text{FFN}(S_L))$$

5. Prediction Layer

$$Z = \text{Pooling}(H_N)$$

$$\hat{r}_{ui} = \sigma(WZ + b)$$

Part 2: Python Code

```
import torch
import torch.nn as nn

class MultiViewTransformer(nn.Module):
    def __init__(self, d_model=128, n_heads=4, num_layers=2, vocab_size=5000):
        super().__init__()
        self.utility_embed = nn.Linear(1, d_model)
        self.text_embed = nn.Embedding(vocab_size, d_model)

        encoder_layer = nn.TransformerEncoderLayer(
            d_model=d_model,
            nhead=n_heads,
            batch_first=True
        )
        self.transformer = nn.TransformerEncoder(encoder_layer,
            num_layers=num_layers)
```

```
self.fc = nn.Linear(d_model, 1)

def forward(self, utility_val, text_tokens):
    u_emb = self.utility_embed(utility_val.unsqueeze(-1))
    t_emb = self.text_embed(text_tokens)
    h0 = torch.cat([u_emb.unsqueeze(1), t_emb], dim=1)
    hN = self.transformer(h0)
    z = hN[:, 0, :]
    return self.fc(z)
```

The mathematical model defines input embeddings, joint representation, and transformer operations with self-attention. The PyTorch code implements this as a MultiViewTransformer class, embedding utility values and text tokens before passing them through a transformer encoder for rating prediction.

QUESTION#03

Paper: A comparative study on the effect of ChatGPT recommendation and AI recommender systems on the formation of a consideration set

H1 Evaluation (200 Words)

H1 proposes that the perceived performance of AI recommender systems, whether traditional systems (H1a) or ChatGPT-generated recommendations (H1b), has a positive influence on user trust in the recommender system. The findings presented in the study support this hypothesis. For traditional AI recommenders, users associate accurate, relevant, and consistent recommendations with system competence. As users perceive improved performance, their trust in the underlying system increases because they view it as reliable and capable of reducing cognitive effort during decision-making. In the case of ChatGPT, perceived performance plays an even more substantial role in trust formation. ChatGPT offers conversational explanations, contextual reasoning, and adaptive responses, which enhance users' perceptions of transparency and intelligence. This encourages a higher level of trust compared to static recommenders. The research results indicate that perceived performance is a strong predictor of trust for both recommender types, with ChatGPT exhibiting a stronger positive effect due to its interactive nature. Overall, H1 is supported. As users perceive better performance, trust in the recommender system increases correspondingly, and the effect is amplified when recommendations are generated through conversational AI such as ChatGPT.

H2 Evaluation (200 Words)

H2 states that the perceived performance of AI recommender systems positively affects trust in the products recommended by the system, both for traditional AI recommenders (H2a) and ChatGPT-generated recommendations (H2b). The results from the study validate this relationship. For traditional recommenders, users tend to project the system's performance onto the quality of the recommended products. When a system consistently provides accurate and relevant suggestions, users interpret this as evidence that the recommended items are dependable and aligned with their preferences. This leads to an increase in trust in the recommended products. With ChatGPT, the influence is even stronger. ChatGPT enhances perceived performance by generating explanations, offering justifications, and tailoring responses to user-specific needs. These conversational interactions create a sense of personalization and competence, which reinforces users' confidence not only in the system but also in the items it suggests. Consequently, perceived performance significantly boosts product-level trust for both systems, with ChatGPT showing a greater impact due to its ability to contextualize and validate recommendations. Therefore, H2 is supported: higher perceived performance leads to increased trust in recommended products, and this effect is more pronounced for ChatGPT-based recommendations.

QUESTION#04

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ACTIVITY #01

Q1. $TN = 119$ $recall = \frac{TP}{TP + FN}$
 $TP = ?$

$FP = 12$ $0.86 = \frac{TP}{TP + 8}$
 $FN = 8$

$recall = 0.86$ $0.86TP + 6.88 = TP$
 $6.88 = 0.14TP$
 $49 = TP$

Q2. $accuracy = \frac{TP + TN}{Sum}$ $precision = \frac{TP}{TP + FP}$ $recall = \frac{TP}{TP + FN} = \frac{2}{2+2} = 0.5$
 $= \frac{2+6}{10} = 0.8$ $= \frac{2}{2+0} = 1$

Q3. $RMSE = \sqrt{\frac{1}{N} \sum (R_{ui} - \hat{R}_{ui})^2}$

$RMSE = \sqrt{\frac{(0.1-1)^2 + (2.5-3)^2 + (3.2-2)^2 + (0.85-2)^2 + (2-4)^2 + (4.5-4)^2 + (0.1-2)^2 + (0.2-2)^2 + (0.5-3)^2}{3 \times 3}}$
 $= 1.562$

Q4.	1	Climate	1 0 0 0 0 0 0 0	climate	1 0 0 0 0 0 0 0
	2	change	0 1 0 0 0 0 0 0	Change	0 1 0 0 0 0 0 0
	3	Generative	0 0 1 0 0 0 0 0	Generative	0 0 1 0 0 0 0 0
	4	Diffusion	0 0 0 1 0 0 0 0	Diffusion	0 0 0 1 0 0 0 0
	5	Models	0 0 0 0 1 0 0 0	Models	0 0 0 0 1 0 0 0
	6	Agents	0 0 0 0 0 1 0 0	Agents	0 0 0 0 0 1 0 0
	7	AI	0 0 0 0 0 0 1 0	AI	0 0 0 0 0 0 1 0
	8	Largechain	0 0 0 0 0 0 0 1	Largechain	0 0 0 0 0 0 0 1
				Change	0 1 0 0 0 0 0 0
				Models	0 0 0 0 1 0 0 0
				AI	0 0 0 0 0 0 1 0

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ACTIVITY #02

PROBLEM (1)

① COSINE SIMILARITY WITH USER 5

u1	0.96
u2	0.85
u3	0.78
u4	0.93
u5	-

$$\cos \theta = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

$$u1 \rightarrow \frac{(4 \times 1) + (3 \times 1) + (4 \times 2) + (2 \times 1)}{\sqrt{4^2 + 3^2 + 4^2 + 2^2} \times \sqrt{1^2 + 1^2 + 2^2 + 1^2}} = 0.96$$

$$u2 \rightarrow \frac{(5 \times 1) + (3 \times 1) + (3 \times 2) + (1 \times 1)}{\sqrt{5^2 + 3^2 + 3^2 + 1^2} \times \sqrt{1^2 + 1^2 + 2^2 + 1^2}} = 0.85$$

$$u3 \rightarrow \frac{(3 \times 1) + (2 \times 1) + (1 \times 2) + (1 \times 1)}{\sqrt{3^2 + 2^2 + 1^2 + 1^2} \times \sqrt{1^2 + 1^2 + 2^2 + 1^2}} = 0.78$$

$$u4 \rightarrow \frac{(3 \times 1) + (1 \times 1) + (3 \times 2) + (1 \times 1)}{\sqrt{3^2 + 1^2 + 3^2 + 1^2} \times \sqrt{1^2 + 1^2 + 2^2 + 1^2}} = 0.93$$

$$\text{PREDICTION} \rightarrow R_4 = \left(\sum_{u=1}^n R_u \right) / n$$

$$\text{taking } k=2 \rightarrow \frac{0.96 \times 5 + 0.93 \times 2}{0.96 + 0.93} = 3.52 \approx \boxed{4}$$

② CORRELATION WITH USER 5 + MEAN CENTERED CORRELATION

u1	3.6	0.48
u2	3.2	0
u3	2	-0.58
u4	2	0.58
u5	1.25	-

$$\text{formula} \rightarrow \frac{\sum_{i=1}^n (x_i - \bar{x}) \times (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \times \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

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$$u_1 \rightarrow \frac{(4-3.6)(1-1.25)^2 + (3-3.6)(1-1.25) + (4-3.6)(2-1.25) + (2-3.6)(1-1.25)}{\sqrt{(4-3.6)^2 + (3-3.6)^2 + (4-3.6)^2 + (2-3.6)^2} \times \sqrt{(1-1.25)^2 + (1-1.25)^2 + (2-1.25)^2 + (1-1.25)^2}}$$

$$= 0.48$$

$$u_2 \rightarrow \frac{(5-3.2)(1-1.25) + (3-3.2)(1-1.25) + (3-3.2)(2-1.25) + (1-3.2)(1-1.25)}{\sqrt{(5-3.2)^2 + (3-3.2)^2 + (3-3.2)^2 + (1-3.2)^2} \times \sqrt{(1-1.25)^2 + (1-1.25)^2 + (2-1.25)^2 + (1-1.25)^2}}$$

$$= 0$$

$$u_3 \rightarrow \frac{(3-2)(1-1.25) + (2-2)(1-1.25) + (1-2)(2-1.25) + (1-2)(1-1.25)}{\sqrt{(3-2)^2 + (2-2)^2 + (1-2)^2 + (1-2)^2} \times \sqrt{(1-1.25)^2 + (1-1.25)^2 + (2-1.25)^2 + (1-1.25)^2}}$$

$$= -0.5$$

$$u_4 \rightarrow \frac{(3-2)(1-1.25) + (1-2)(1-1.25) + (3-2)(2-1.25) + (1-2)(1-1.25)}{\sqrt{(3-2)^2 + (1-2)^2 + (3-2)^2 + (1-2)^2} \times \sqrt{(1-1.25)^2 + (1-1.25)^2 + (2-1.25)^2 + (1-1.25)^2}}$$

$$= 0.58$$

~~Adjusted cosine sim~~

$$\text{PREDICTION} \rightarrow \hat{R}_{ij} = \frac{\bar{r}_i + \sum_b GN \text{sim}(r_i, r_b) * (r_b - \bar{r}_b)}{\sum_b GN \text{sim}(r_i, r_b)}$$

$$= 1.25 + \frac{0.48(5-3.6) + 0.58(2-2)}{0.48 + 0.58}$$

$$= 1.88 \approx \boxed{2}$$

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④ ADJUSTED COSINE

41	3.6	0.48
42	3.2	0
43	2	-0.5
44	2	0.58
45	1.25	-

from previous part.

$$\text{PREDICTION} = \frac{0.48 \times 5 + 0.58 \times 2}{0.48 + 0.58} = 3.36 \approx \boxed{3}$$

PROBLEM ②

$$TF = \frac{\text{freq.}}{\text{max of any other term}}$$

$$IDF = \lg \left(\frac{N}{n_k} \right)$$

	AT	RS	CP
R1	1.1034	0.9063	0.0313
R2	0.4839	2.0667	0.0806
R3	0.5510	1.8148	0.0816
R4	0.5357	1.8667	0.2143
R5	0.3333	3	0.1429
R6	0.3333	3	0.2000

AT	$\lg \left(\frac{60000}{2600} \right) = 1.3632$
RS	$\lg \left(\frac{60009}{1400} \right) = 1.6320$
CP	$\lg \left(\frac{60009}{900} \right) = 1.8239$

(divide by magnitude)

	AT	RS	CP
R1	0.7128	0.7009	0.0270
R2	0.1918	0.9805	0.0427
R3	0.2456	0.9682	0.0486
R4	0.2314	0.9650	0.1238
R5	0.0913	0.4443	0.0529
R6	0.0913	0.4443	0.0733

	TF-IDF _{AT}	RS	CP
R1	1.5042	1.4791	0.0371
R2	0.6598	3.3729	0.1470
R3	0.7512	2.9618	0.1488
R4	0.7304	3.0465	0.3909
R5	0.4544	4.8960	0.2608
R6	0.4544	4.8960	0.3648

$$\text{magnitude R1} \rightarrow \sqrt{1.5042^2 + 1.4791^2 + 0.0371^2} = 2.1104$$

do for all

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COSINE SIMILARITY WITH R1
(Since normalized, using dot product)

$$\begin{aligned} R2 \rightarrow & (0.7128 \times 0.1918) + (0.7009 \times 0.9805) + (0.0270 \times \\ & 0.0477) \\ & = 0.8251 \end{aligned}$$

$$\begin{aligned} R3 \rightarrow & (0.7128 \times 0.2456) + (0.7009 \times 0.9682) + (0.0270 \times \\ & 0.0486) \\ & = 0.8551 \end{aligned}$$

$$\begin{aligned} R4 \rightarrow & (0.7128 \times 0.2314) + (0.7009 \times 0.9650) + (0.0270 \times \\ & 0.1238) \\ & = 0.8447 \end{aligned}$$

$$\begin{aligned} R5 \rightarrow & (0.7128 \times 0.0923) + (0.7009 \times 0.9943) + (0.0270 \times \\ & 0.0529) \\ & = 0.7643 \end{aligned}$$

$$\begin{aligned} R6 \rightarrow & (0.7128 \times 0.0922) + (0.7009 \times 0.9930) + \\ & (0.0270 \times 0.0740) \\ & = 0.7637 \end{aligned}$$

using $k=2$, ⁽¹⁾ R3 & ⁽²⁾ R4 recommended

ACTIVITY#03

QUESTION#01 – CF RESEARCH PAPER

1. What is the main goal of a recommender system?

To alleviate information overload by filtering data and identifying relevant items from large item sets. (Page 1)

2. What are the two main types of collaborative filtering approaches?

Memory-based and model-based approaches. (Page 1)

3. What is the key difference between memory-based and model-based CF?

Memory-based uses the rating matrix directly; model-based learns a model from data (e.g., MF). (Page 1)

4. Which algorithm is the proposed approach based on?

Naïve Bayes Classifier (NBC). (Page 2)

5. What assumption does Naïve Bayes make about variables?

Conditional independence between all variable pairs. (Page 2)

6. What three approaches were proposed?

User-based, item-based, and hybrid. (Page 2)

7. What does the parameter α represent?

A smoothing parameter to avoid zero probabilities. (Page 2)

8. How is the predicted rating $\hat{r}_{(u,i)}$ computed?

By selecting the rating y that maximizes $P(r_{u,i} = y)$. (Page 3)

9. What does the reliability measure $l_{u,i}$ indicate?

The confidence level of the system in its prediction. (Page 3)

10. Which datasets were used in the experiments?

MovieLens, FilmTrust, Yahoo, BookCrossing. (Page 5)

11. What is the purpose of using nDCG?

To measure ranking accuracy. (Page 7)

12. Which dataset showed the best MAE improvement with NBCF?

BookCrossing (MAE = 1.48323). (Page 8)

13. What does NBCF stand for?

Naïve Bayes Collaborative Filtering.

14. Which baseline methods were used for comparison?

BNMF, GGM, INBM, Bi-CF, NMF. (Pages 5–6)

15. How does the hybrid NBCF approach combine evidence?

Through a weighted product based on strength of user-based and item-based evidence. (Page 3)

16. What is the computational complexity of NBCF?

$O(U \cdot \bar{I} \cdot (\bar{I} + \bar{U}))$. (Page 11)

17. How can the model explain its recommendations?

By showing influential items/users used in the probability estimation. (Page 11)

18. What happens to accuracy as reliability increases?

The percentage of perfect predictions (error = 0) increases. (Page 10)

19. What future improvements were suggested?

Integration of attributes, social info, context, and parallelization. (Page 12)

20. How is $\hat{r}_{(u,i)}$ obtained in NBCF?

$\operatorname{argmax}_y P(r_{u,i} = y)$. (Page 3)

21. What does $l_{u,i}$ indicate?

Confidence in the predicted rating. (Page 3)

22. What does reliability = 0.55 correspond to?

Approximately 70% perfect predictions. (Page 10)

23. What is the time complexity of NBCF?

$O(U \cdot \bar{I} \cdot (\bar{I} + \bar{U}))$. (Page 11)

24. What do $\bar{U}$ and $\bar{I}$ represent?

$\bar{U}$ = average users per item; $\bar{I}$ = average items per user. (Page 11)

25. How does NBCF reduce computational cost?

By memorizing intermediate probability values. (Page 5)

26. What metrics were used to measure accuracy?

MAE, Precision, Recall, nDCG. (Page 7)

27. What does MAE measure?

Mean absolute error between predicted and actual ratings. (Page 7)

28. What does nDCG evaluate?

Ranking quality. (Page 7)

29. What is the role of reliability?

To filter out low-confidence predictions and boost accuracy. (Page 3)

30. How many users are in the MovieLens dataset?

6,040 users. (Page 6)

QUESTION#02 – KBRs RESEARCH PAPER

1. What is the primary goal of a recommender system?

To identify relevant items and reduce decision complexity. (Page 1)

2. How do recommender systems reduce complexity?

By filtering information and presenting personalized suggestions. (Page 1)

3. Difference between single-shot and conversational recommenders?

Single-shot uses fixed user profiles; conversational engages in dialog-based preference elicitation. (Page 1)

4. Explain collaborative filtering (CF).

Uses preferences of similar users (nearest neighbors). (Page 1)

5. What is the nearest neighbor concept?

Users with similar preference patterns. (Page 1)

6. What is the cold-start problem?

Lack of initial ratings leads to poor CF recommendations. (Page 2)

7. What is content-based filtering (CBF)?

Recommends items similar to those a user previously liked, using item attributes. (Page 2)

8. Role of item knowledge in CBF?

Used to build profiles and compute similarity. (Page 2)

9. Which items suit knowledge-based recommenders?

High-involvement, complex items like real estate or financial products. (Page 2)

10. Two main types of KBR?

Constraint-based and case-based. (Page 2)

11. What is preference elicitation?

Collecting user requirements interactively. (Page 2)

12. Difference between table-based and constraint-based representation?

Table-based enumerates all options; constraint-based defines rules governing valid solutions. (Pages 4–5)

13. What problem does constraint-based recommendation solve?

Constraint satisfaction problems (CSP). (Page 5)

14. How do constraints define valid configurations?

Through logical rules that items must satisfy. (Page 5)

15. What is a case base?

Stored set of cases/items used to retrieve similar items. (Page 5)

16. How is similarity computed in case-based recommenders?

Using attribute-level similarity metrics. (Page 6)

17. Define the interest confidence value (ICV).

Average similarity between a new item and previously liked items. (Pages 5–6)

18. What are attribute-level similarity metrics?

MIB, LIB, EIB per attribute. (Page 6)

19. Difference between MIB and LIB?

MIB: higher values better; LIB: lower values better. (Page 6)

20. What is the EIB metric?

Exact attribute match gives highest similarity. (Page 6)

21. What is a critiquing-based recommender?

A system where users critique an item to refine recommendations. (Page 6)

22. Explain unit critiquing.

One attribute critique at a time. (Pages 6–7)

23. Purpose of compound critiques?

Allow multiple critiques simultaneously to reduce interaction cycles. (Page 7)

24. How are conflict sets identified?

As minimal sets of user requirements inconsistent with constraints. (Page 9)

25. What is a diagnosis?

Minimal requirement set to remove to restore consistency. (Page 9)

26. Strategies to handle inconsistent preferences?

Conflict detection, diagnosis, relaxation, compromise. (Pages 6, 9)

27. How are ranking/utility functions used?

Assign scores based on features and interest dimensions. (Page 9)

28. What is a CB-REC task?

A CSP involving variables, constraints, and user requirements. (Page 8)

29. How does the system respond when no solution exists?

Suggests diagnoses or compromise solutions. (Page 9)

30. Key application domains for KBR systems?

Financial services, real estate, software, digital cameras, survey tools. (Pages 2, 4, 7–10)

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ACTIVITY #05

PROBLEM (1)

CONCATENATED VECTORS

$$(U_1, I_1) = [0.1, 0.3, 0.5, 0.2]$$

$$(U_2, I_2) = [0.2, 0.4, 0.4, 0.6]$$

$$(U_3, I_3) = [0.1, 0.3, 0.4, 0.6]$$

$$(U_4, I_4) = [0.2, 0.4, 0.5, 0.2]$$

HIDDEN LAYER CALCS

$$z = Wx + b$$

$$z_1 = (0.2 \times 0.1 + 0.1 \times 0.3 + 0.4 \times 0.5 + 0.3 \times 0.2) + 0.1$$
$$= 0.41$$

$$z_2 = (0.5 \times 0.1 + 0.3 \times 0.3 + 0.2 \times 0.5 + 0.6 \times 0.2) + 0.1$$
$$= 0.76$$

$$h_1 = \delta(0.41) = \frac{1}{1 + e^{-0.41}} = 0.6011$$

$$h_2 = \delta(0.76) = \frac{1}{1 + e^{-0.76}} = 0.6813$$

$$h = [0.6011, 0.6813]$$

$$z = Wx + b$$

$$z_1 = (0.2 \times 0.2 + 0.4 \times 0.4 + 0.4 \times 0.4 + 0.3 \times 0.6) + 0.1$$
$$= 0.52$$

$$z_2 = (0.5 \times 0.2 + 0.3 \times 0.4 + 0.2 \times 0.4 + 0.6 \times 0.6) + 0.4$$
$$= 1.06$$

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$$h_1 = \delta(0.52) = \frac{1}{1+e^{-0.52}} = 0.6271$$

$$h_2 = \delta(1.06) = \frac{1}{1+e^{-1.06}} = 0.7426$$

$$h = [0.6271, 0.7426]$$

$$z = w_0 + b \quad u_{1I2}$$

$$z_1 = (0.2 \times 0.1 + 0.1 \times 0.3 + 0.4 \times 0.4 + 0.3 \times 0.6) + 0.1 = 0.49$$

$$z_2 = (0.5 \times 0.1 + 0.3 \times 0.3 + 0.3 \times 0.4 + 0.6 \times 0.6) + 0.1 = 0.98$$

$$h_1 = \delta(0.49) = \frac{1}{1+e^{-0.49}} = 0.6201$$

$$h_2 = \delta(0.98) = \frac{1}{1+e^{-0.98}} = 0.7271$$

$$h = [0.6201, 0.7271]$$

$$z = w_0 + b \quad u_{2I1}$$

$$z_1 = (0.2 \times 0.2 + 0.1 \times 0.4 + 0.4 \times 0.5 + 0.3 \times 0.2) + 0.1 = 0.44$$

$$z_2 = (0.5 \times 0.2 + 0.3 \times 0.4 + 0.2 \times 0.5 + 0.6 \times 0.2) + 0.1 = 0.84$$

$$h_1 = \delta(0.44) = \frac{1}{1+e^{-0.44}} = 0.6083$$

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$$h_2 = \delta(0.84) = \frac{1}{1 + e^{-0.84}} = 0.6985$$

$$h = [0.6083, 0.6985]$$

OUTPUT LAYER CALCS

U1I1

$$\hat{y} = (-1.2 \times 0.6011 + 0.9 \times 0.6813) + 0.2$$

$$= 0.0919$$

U2I1

$$\hat{y} = (-1.2 \times 0.6271 + 0.9 \times 0.7426) + 0.2$$

$$= 0.1158$$

U2I2

$$\hat{y} = (-1.2 \times 0.6201 + 0.9 \times 0.7271) + 0.2$$

$$= 0.1103$$

U2I3

$$\hat{y} = (-1.2 \times 0.6083 + 0.9 \times 0.6985) + 0.2$$

$$= 0.0987$$

PROBLEM 2

$$U1I1 \rightarrow (1 - 0.0919)^2 = 0.8246$$

$$\text{sum} = 1.6283$$

$$U2I2 \rightarrow (1 - 0.1158)^2 = 0.7818$$

$$\text{MSE} = \frac{1.6283}{4} = 0.4071$$

$$U1I2 \rightarrow (0 - 0.1103)^2 = 0.0122$$

$$U2I1 \rightarrow (0 - 0.0987)^2 = 0.0097$$

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PROBLEM ③
HIDDEN LAYER CALCS
ReLU = max(0, z)

$$\begin{aligned} \text{U2I1} &\rightarrow z_1 = 0.41 \\ h_1 &= \max(0, 0.41) = 0.41 \\ z_2 &= 0.76 \\ h_2 &= \max(0, 0.76) = 0.76 \end{aligned}$$

$$\begin{aligned} \text{U2I2} &\rightarrow z_1 = 0.52 \\ h_1 &= \max(0, 0.52) = 0.52 \\ z_2 &= 1.06 \\ h_2 &= \max(0, 1.06) = 1.06 \end{aligned}$$

$$\begin{aligned} \text{U1I2} &\rightarrow z_1 = 0.49 \\ h_1 &= \max(0, 0.49) = 0.49 \\ z_2 &= 0.98 \\ h_2 &= \max(0, 0.98) = 0.98 \end{aligned}$$

$$\begin{aligned} \text{U2I1} &\rightarrow z_1 = 0.44 \\ h_1 &= \max(0, 0.44) = 0.44 \\ z_2 &= 0.84 \\ h_2 &= \max(0, 0.84) = 0.84 \end{aligned}$$

OUTPUT LAYER

$$\text{U1I1} \rightarrow \hat{y} = (-1.2 \times 0.41 + 0.9 \times 0.76) + 0.2 = 0.3970$$

$$\text{U2I2} \rightarrow \hat{y} = (-1.2 \times 0.52 + 0.9 \times 1.06) + 0.2 = 0.5300$$

$$\text{U1I2} \rightarrow \hat{y} = (-1.2 \times 0.49 + 0.9 \times 0.98) + 0.2 = 0.4940$$

$$\text{U2I1} \rightarrow \hat{y} = (-1.2 \times 0.44 + 0.9 \times 0.84) + 0.2 = 0.4280$$

Date: _____

LOSS CALCS

$$u1I1 \rightarrow (1 - 0.3920)^2 = 0.3697$$

$$\text{sum} = 1.0178$$

$$u2I2 \rightarrow (1 - 0.5300)^2 = 0.2209$$

$$\text{MSE} = \frac{1.0178}{4} = 0.2545$$

$$u1I2 \rightarrow (0 - 0.4940)^2 = 0.2440$$

$$u2I1 \rightarrow (0 - 0.4280)^2 = 0.1832$$

Relu gives higher predicted ratings (closer to .5 range)
and lower MSE than sigmoid because it preserves linearity
for pos inputs